MIL-D-11243A(Cm1C)
23 February 1961
SUPERSEDING
MIL-G-11243(Cm1C)
23 June 1951
(See section 6)

MILITARY SPECIFICATION

DEXTROSE, MONOHYDRATE, TECHNICAL (d-GLUCOSE)

1. SCOPE

1.1 Scope. This specification covers two grades of technical monohydrate dextrose for use in biological media.

1.2 Classification. The dextrose shall be of the following grades as specified (see 6.1):

Grade A - Low heavy metals content.
Grade B - Low arsenic content.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-S-48 - Sacks, Shipping, Paper.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Copies of the Pharmacopeia of the United States (USP) may be obtained from The Mack Publishing Co., Easton, Pa.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Appearance. The dextrose shall be a granular or crystalline white powder.

3.2 Chemical and physical properties. The chemical and physical properties of the dextrose shall conform to the requirements specified in table I when tested in accordance with the test paragraphs referenced therein:

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Table I. - Chemical and Physical Properties

Property	Requirements		Test paragraph
	Grade A	Grade B	
Loss on drying at 100°C (% min.)	6.0	6.0	4.6.1
(% max.)	10.0	10.0	
Specific rotation $[\alpha]_D^{25}$ (min.)	∠ 52.5°	∠ 52.5°	4.6.2
(max.)	∠ 53.0°	∠ 53.0°	
Acidity (as hydrochloric acid) (% max.)	0.005	0.01	4.6.3
Sulfates (% max.)	0.03	0.03	4.6.4
Heavy metals (expressed as lead) (% max.)	0.0005	0.001	4.6.5
Arsenic (% max.)	0.0005	0.0001	4.6.6
Soluble starch, dextrin or sulfites	To pass test	To pass test	4.6.7
Chlorides (% max.)	0.02	0.05	4.6.8

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

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4.2.1 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the inspection procedures of this specification will govern.

4.2.2 Supplemental evidence. The supplier shall provide evidence acceptable to the contracting officer that the requirements of section 5 have been satisfied. (See 6.2.)

4.3 Lotting. A lot shall consist of the dextrose produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials, provided the process is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.2).

4.4 Sampling.

4.4.1 For nondestructive examination. Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.4.2 For test. Sample containers shall be taken at random from each lot as follows:

Lot size	Sample size
1	1
2- 275	2
276- 545	3
546- 900	4
901-1345	5
1346-1875	6
1876-2500	7

Sample size for lots exceeding 2500 containers shall be calculated using the following formula:

$$n = 0.15 \sqrt{N}$$

where: n = Sample size.
N = Lot size.

A specimen shall be removed from each container in the sample by means of a clean, dry thief, and placed in a clean, dry container labeled to identify the lot and container from which the specimen was taken. Each specimen shall be examined for compliance with 3.1 and be tested as specified in 4.6.

4.5 Examination. Sample filled containers of dextrose shall be examined for compliance with levels A and B packing in accordance with the classification of defects and Standard MIL-STD-105.

4.5.1 Classification of defects.

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4.5.1.1 Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.5 percent defective
101	Containers not as specified
102	Containers damaged
103	Closure incorrect
104	Marking incorrect, illegible or incomplete

4.6 Tests. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. When applicable, blank determinations shall be conducted in parallel with the tests using the same quantities of reagents, and corrections applied when significant. All weighings shall be made to an accuracy of one milligram (mg.). Tests shall be conducted as follows:

4.6.1 Loss on drying at 100°C. Weigh 10 ± 1 g of the specimen (see 4.4.2) in a previously tared weighing dish. Dry the dextrose to constant weight in an oven at a temperature of 100° ± 2°C. (The dried dextrose may be saved for the test specified in 4.6.2.) Calculate the percent loss on drying as follows:

$$\text{Loss on drying at 100°C (percent)} = \frac{100A}{W}$$

where: A = Loss of weight on drying.
W = Weight of dextrose used.

4.6.2 Specific rotation. Prepare approximately 100 milliliters (ml) of an aqueous solution containing 10,000 mg of the dried dextrose (see 4.6.1) per 100.0 ml of solution. Mix two-tenths ml of dilute ammonium hydroxide (395 ml of concentrated ammonium hydroxide diluted to 1 liter) with the solution. Fill a 200-millimeter polarimeter tube with the alkaline dextrose solution. Read the rotation, at a temperature of 25.00° ± 0.25°C, by means of a polarimeter using a sodium vapor light for illumination. Calculate the specific rotation as follows:

$$\text{Specific rotation } [\alpha]_D^{25} = \frac{5A}{D}$$

where: A = Observed rotation in degrees.

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4.6.3 Acidity. Weigh 5 ± 1 g of the specimen and dissolve it in 50.0 ml of carbon dioxide free water. Add three drops of phenolphthalein, and titrate the mixture with 0.02N sodium hydroxide to a faint pink color. Calculate the acidity (as hydrochloric acid) as follows:

$$\text{Acidity (as percent hydrochloric acid} = \frac{3.65\ V\ N}{W}$$

where: V = Volume of sodium hydroxide (in ml) used.
N = Normality of sodium hydroxide.
W = Weight (in g) of dextrose used.

4.6.4 Sulfates.

4.6.4.1 Solutions.

4.6.4.1.1 Barium chloride solution. Dissolve 12.0 g of barium chloride dihydrate in 100.0 ml of water.

4.6.4.1.2 Potassium sulfate standard solution. Dissolve 0.1814 g of potassium sulfate in water and dilute to 1,000.0 ml with water.

4.6.4.1.3 Test solution. Dissolve 0.500 g of the specimen in 20.0 ml of water and add 1.0 ml of 1.0N hydrochloric acid. Pour the solution into an identified matched comparison tube.

4.6.4.1.4 Standard solution. Pipette 1.50 ml of potassium sulfate standard solution into 20.0 ml of water and add 1.0 ml of 1.0N hydrochloric acid. Pour the solution into a second identified matched comparison tube.

4.6.4.2 Procedure. Place 2.0 ml of barium chloride solution into each of the comparison tubes containing the test solution and the standard solution, and thoroughly mix the contents of the tubes. Using a comparator, view the mixtures lengthwise through the comparison tubes to determine compliance with 3.2. Report the lot as meeting the requirement for sulfates if the test solution is not more turbid than the standard solution.

4.6.5 Heavy metals (as lead).

4.6.5.1 Solutions.

4.6.5.1.1 Hydrochloric acid solution. Dilute 236 ml of concentrated hydrochloric acid to 1 liter with water.

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4.6.5.1.2 Bromine solution. Agitate 2 to 3 ml of bromine with approximately 100 ml of water in a glass-stoppered bottle having the stopper lubricated with petrolatum. Store in a cool, dark place.

4.6.5.1.3 Acetic acid solution. Dilute 60 ml of glacial acetic acid to 1 liter with water.

4.6.5.1.4 Hydrogen sulfide solution. Pass hydrogen sulfide through cold water until a saturated solution is obtained.

4.6.5.1.5 Lead solutions.

4.6.5.1.5.1 Stock lead solution. Dissolve 0.160 g of lead nitrate in 100 ml of water, add 1 ml of concentrated nitric acid and dilute to 1 liter.

4.6.5.1.5.2 Standard lead solution. Dilute 10.0 ml of the stock lead solution to 100.0 ml with water. Prepare this solution immediately before use.

4.6.5.2 Qualitative test. Dissolve approximately 1 g of the specimen in 10 ml of water, add approximately 2 ml of hydrogen sulfide solution and observe the mixture for any change in color or turbidity. If no change occurs, report the lot as meeting the requirement in 3.2 If a change in turbidity or color is observed, conduct the quantitative test.

4.6.5.3 Quantitative test. Thoroughly char 5.00 g of the specimen in a porcelain dish, add approximately 30 ml of hydrochloric acid solution and 3 ml of bromine solution, cover the dish, and boil gently for approximately 10 minutes. Filter the solution, and wash the filter and residue with approximately 25 ml of hot water. Evaporate the filtrate to approximately 15 ml, cool, and add ammonia solution, dropwise, until the solution is neutral to litmus paper. Add 2 ml of acetic acid solution, and dilute to 25 ml with water. (Filter, if necessary, to obtain a clear solution.) Transfer the solution to a 50-ml Nessler tube identified with the specimen. Prepare a standard for grade A dextrose by mixing 2.5 ml of standard lead solution (use 5.0 ml for grade B dextrose), 2 ml of acetic acid solution and 25 ml of water, and pour it into a second 50-ml Nessler tube identified as the standard. Add 10.0 ml of hydrogen sulfide solution to each of the Nessler tubes, thoroughly agitate the contents of the tubes, and allow them to stand for approximately 10 minutes. View the contents of the tubes lengthwise in a comparator over a white surface. Report the lot as meeting the requirement for heavy metals if the color of the test solution is no darker than the standard solution.

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4.6.6 Arsenic.

4.6.6.1 Solutions.

4.6.6.1.1 Dilute sulfuric acid. Slowly pour 57 ml of concentrated sulfuric acid into approximately 500 ml of water with constant agitation. Cool the solution and dilute to 1,000 ml.

4.6.6.1.2 Stannous chloride solution. Dissolve 1.50 g of stannous chloride in 10.0 ml of water made slightly acid with hydrochloric acid.

4.6.6.1.3 Bromine solution. Use the bromine solution prepared as specified in 4.6.5.1.2.

4.6.6.2 Procedure. Prepare an aqueous solution containing 1.500 g of the specimen per 5.0 ml of solution. To 5.0 ml of this solution add 5.0 ml dilute sulfuric acid, and 1 ml of bromine solution. Heat the mixture for 5 minutes on a water bath. Add 5.00 g of potassium iodide and five drops of stannous chloride solution, and cool the solution to room temperature. Determine the percent arsenic using the Gutzeit procedure specified in the Pharmacopeia of the United States.

4.6.7 Soluble starch, dextrin or sulfites.

4.6.7.1 Procedures.

4.6.7.1.1 Soluble starch or dextrin. Dissolve a 5.0 g portion of the specimen in 50 ml of water. Boil the solution for one minute, cool to room temperature, and add one drop of 0.1N iodine. Observe any change of color in the solution. Report the lot as meeting the requirement for soluble starch and dextrin if no red or blue coloration is observed.

4.6.7.1.2 Sulfites. Add four drops of a 1-percent soluble starch solution in water to the solution obtained in 4.6.7.1.1. Observe the color of the solution. Report the lot as meeting the requirement for sulfites if a blue color is observed.

4.6.8 Chlorides.

4.6.8.1 Solutions.

4.6.8.1.1 Standard chloride solution. Dissolve 0.165 g of sodium chloride in water and make up to 1,000.0 ml. Thoroughly mix the solution.

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4.6.8.1.2 Silver nitrate solution. Dissolve 2.0 g of silver nitrate in approximately 100 ml of water.

4.6.8.1.3 Standard solution. Pipette 1.0 ml of standard chloride solution (see 4.6.8.1.1) into approximately 20 ml of water and add approximately 1 ml of concentrated nitric acid.

4.6.8.1.4 Test solution. For grade A dextrose, dissolve 0.500 g of the specimen (0.200 g for grade B dextrose) in approximately 20 ml of water and add approximately 1 ml of concentrated nitric acid. (Filter if necessary to obtain a clear solution.)

4.6.8.2 Procedure. Pour the standard solution and the test solution into separate identified matched comparison tubes, add approximately 1 ml of silver nitrate solution to each tube, and thoroughly mix the contents of the tubes. View the contents of the tubes lengthwise through the tubes to determine their turbidities. Report the sample as meeting the requirement for chlorides if the test solution is not more turbid than the standard solution.

4.7 Acceptance/rejection criteria. If any specimen fails to meet the requirements of section 3, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. Dextrose in 50-pound quantities shall be packed in paper shipping sacks conforming to type III or type IV, construction 31X, Specification UU-S-48. The sacks shall be fabricated from class B paper and shall have tuck-in-sleeves and dipped ends.

5.1.2 Level B. Dextrose in 50- or 100-pound quantities, as required, shall be packed as specified in 5.1.1 except that the sacks for 50-pound quantities shall conform to construction 3 L/W and sacks for 100-pound quantities shall conform to construction 12 L/W.

5.1.3 Level C. Fifty or one hundred pounds, as required, of dextrose shall be packed in a manner to insure carrier acceptance with safe delivery at first destination for immediate use. Containers shall be in accordance with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation. Containers for like quantities shall be of uniform size, shape, and material.

5.2 Marking. In addition to any special marking required by the

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contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Grade.
(c) Level of packing required.
(d) Net weight of contents in shipping container.
(e) Supplemental evidence required.

6.2 Batch. A batch is defined as that quantity of material which has been subjected to some physical mixing process intended to make the final product substantially uniform.

6.3 Supersession data. This specification supersedes Purchase Description 198-2-13B, dated 27 October 1959.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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